Title: Understanding Technology Foundations – Week 1

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Course: Risk Management Framework

Class: 1

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Part A – Short Answer (Concept Mastery)

1. **Explain in your own words what a computer system is.**

   A computer system is a combination of hardware and software that work together to process data and perform tasks. Hardware includes physical components such as the CPU, memory, and storage. Software includes programs that instruct the computer on what actions to perform.

2. **Describe the three core components of computing: Input, Processing, Output. Provide one real-world example for each.**

Component	**Description**	**Example**
Input	Data entered into a computer	Typing a username and password with a keyboard to log in.
Processing	The computer processes instructions	The system verifying your password against a user database.
Output	The computer produces results	Displaying “Login successful” and loading your dashboard.

3. **What is the difference between hardware and software?**

Type	**Description**	**Example**
Hardware	Physical components of a computer	Keyboard, CPU, RAM, monitor, hard drive.
Software	Programs that run on the computer	Windows, Applications, Linux, browsers, Office, security tools.

4. **Explain what a router does in a network.**

   A router connects different networks and directs data between them. It sends internet traffic between devices in a local network and external networks.

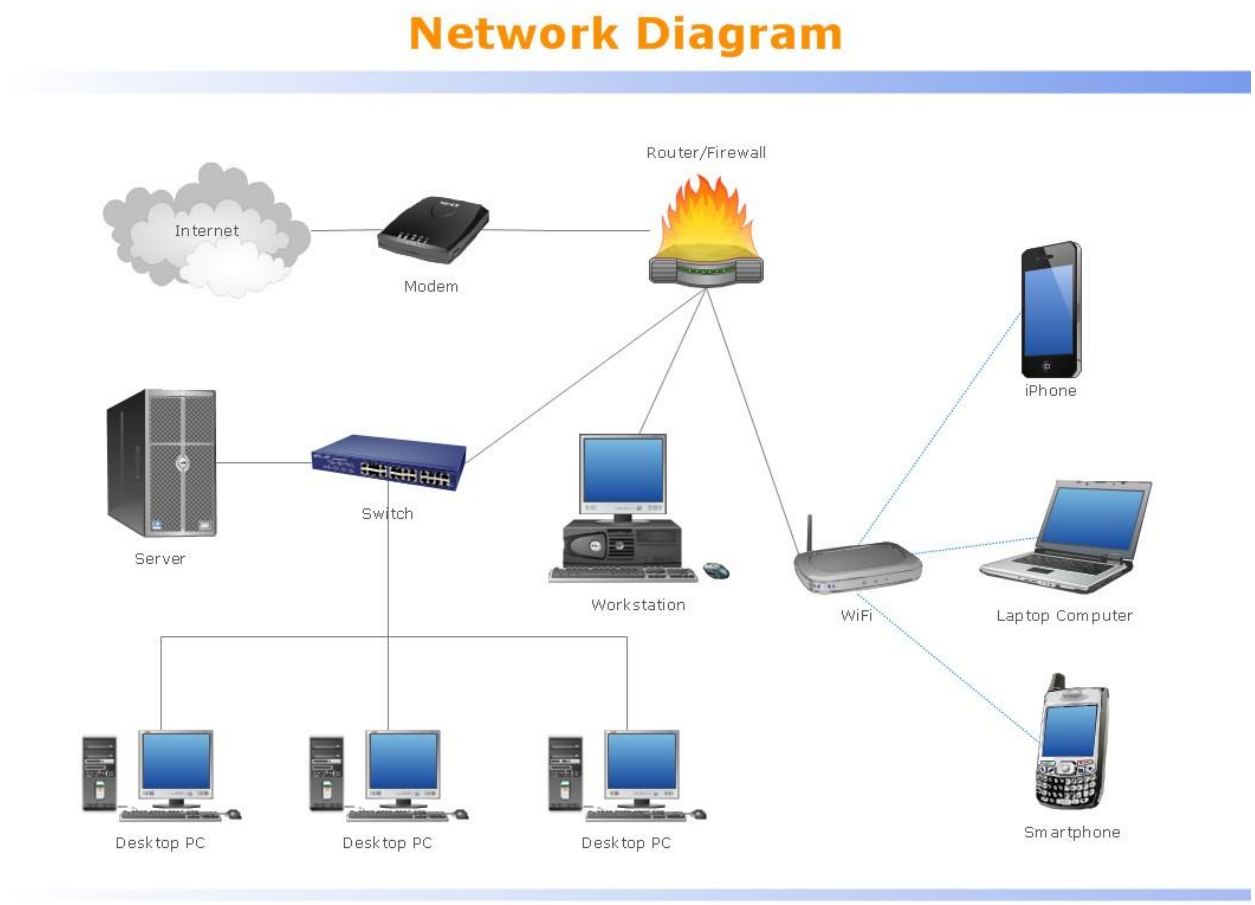

Diagram Description

The diagram shows how internet connectivity flows through a network. The internet connects to a modem, which sends the connection to a router/firewall. The router distributes the connection to a switch for wired devices and WiFi for wireless devices such as laptops and smartphones. Wired devices like servers and desktop computers connect through the switch.

5. What is the difference between a modem and a router?

Device	Function
Modem	Connects a network to the internet service provider
Router	Distributes the internet connection to devices

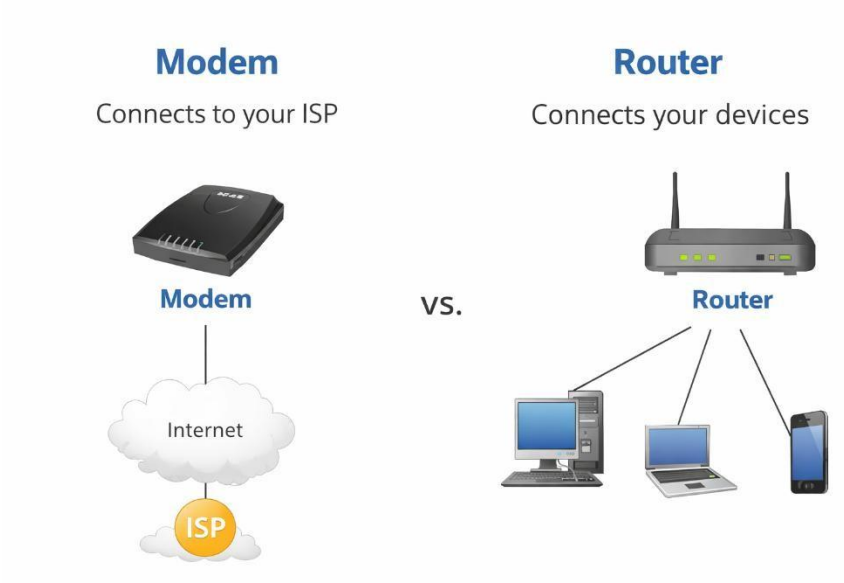

Diagram Description

The diagram compares a modem and a router.

The modem connects the network to the Internet Service Provider (ISP) and provides internet access. The router distributes the internet connection to multiple devices such as computers, laptops, and smartphones within the local network.

6. What is a network switch, and how is it different from a router?

Device	OSI layer & scope	What it does	How it makes decisions	Typical use in a network
Switch	Layer 2, works inside a single LAN	Connects multiple devices in the same network	Uses MAC addresses in frames	Builds the internal LAN (access/distribution layer)
Switch	Layer 3, connects different networks/LANs	Connects LANs to other networks (e.g., internet/WAN)	Uses IP addresses in packets	Edge device between LAN and ISP, enforces NAT/firewall

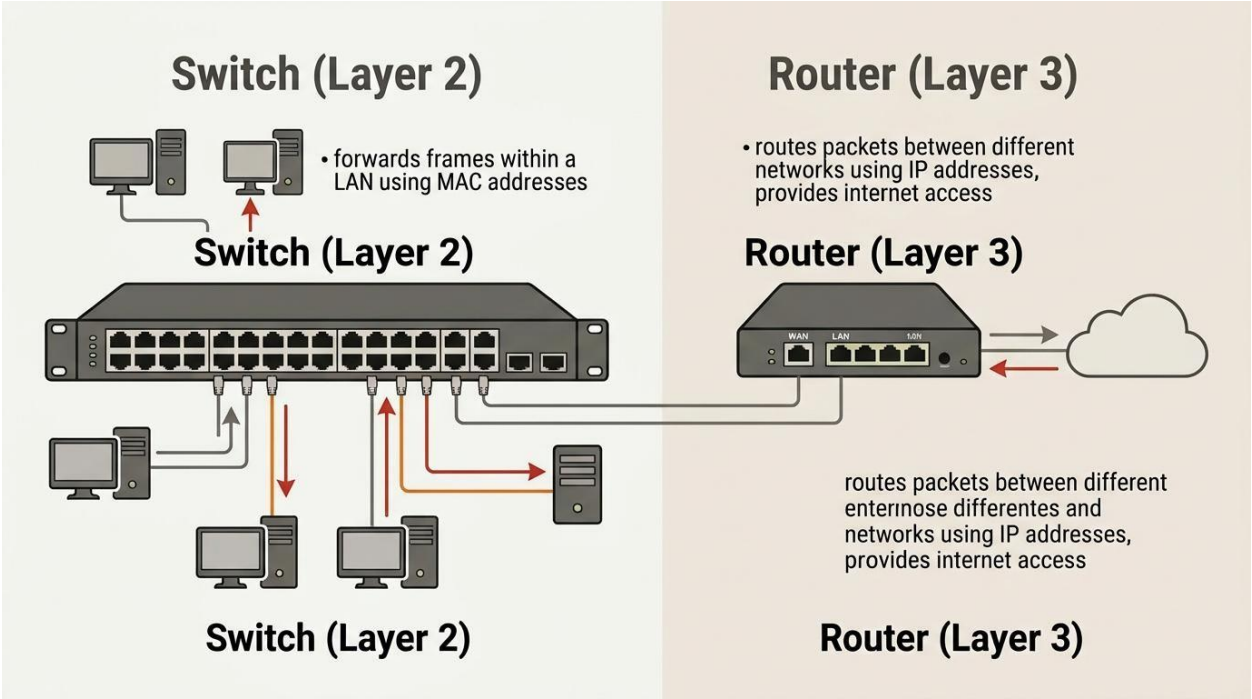

Diagram Description

The diagram shows a switch connecting multiple devices inside a single LAN using MAC addresses, and a router sitting in front, connecting that LAN to the internet and other networks using IP addresses.

7. Explain what storage means in computing. List and briefly describe: HDD, SSD, Cloud storage.

Type	Description	Key traits
HDD	Magnetic disks with spinning platters.	Cheap, high capacity; mechanical → vibration/shock risk.
SSD	Flash memory chips, no moving parts.	Fast, durable; encrypt at rest (e.g., BitLocker).
Cloud	Remote servers accessed via internet.	Scalable; use MFA, encryption in transit/rest.

8. What is a data center? Why are data centers important to businesses?

It's a secure facility housing servers, storage, and networking for data processing/storage

Why Important to Businesses?

Reason	Business Impact
Data storage/backup	Ensures availability, redundancy (RAID, offsite).
Scalability	Handles growth without downtime.

Security	Physical (biometrics), network controls (firewalls).
Disaster recovery	RTO/RPO compliance, business continuity.

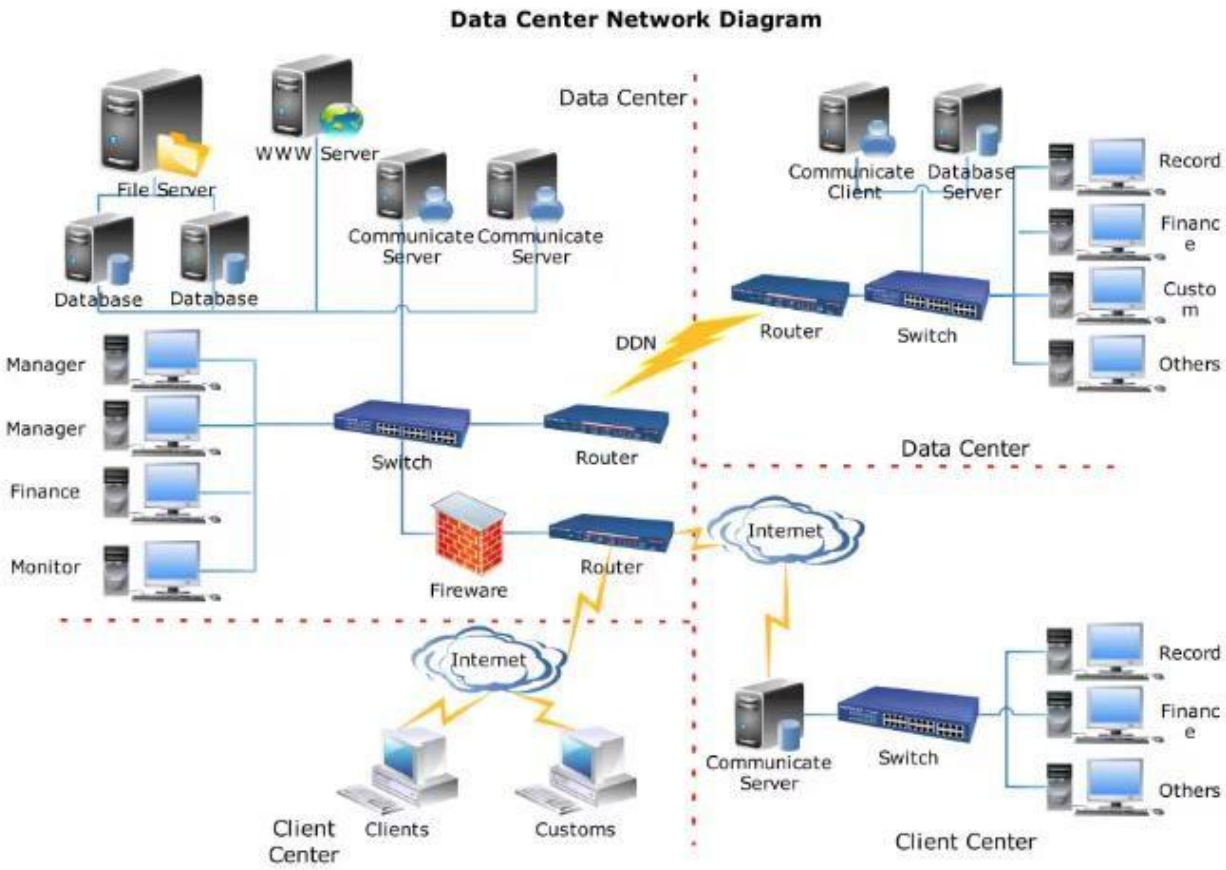

Diagram Description

The diagram shows users connecting through the internet to a data center network. Inside the data center are servers, storage systems, and networking equipment that process applications and store business data.

9. **Define cloud computing in your own words.**

Cloud computing is using computers and servers on the internet to store data, run applications, and access services instead of relying only on your personal computer.

10. **Explain the three cloud service models: IaaS, PaaS, SaaS. Provide one real-world example of each.**

Model	What it provides	You manage	Example

IaaS	Virtual infrastructure (servers, storage, networks)	OS, apps, data	AWS EC2
PaaS	Platform + tools for app development	Apps, data	Google App Engine
SaaS	Fully managed software apps	Just data/users	Microsoft 365

Part B – Application & Critical Thinking

11. **Draw a simple diagram showing how a user accesses a website. Include: User device, Router, ISP, Server, Data center. Label everything clearly.**

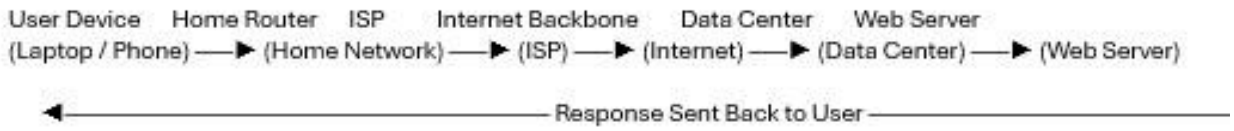

Diagram Description

The user device sends a request through the home router and ISP to the internet backbone. The request reaches a web server located in a data center, which processes the request and sends the response back to the user.

12. **Explain what happens behind the scenes when you type 'google.com' into a browser.**

    1. The browser sends a request for the website.
    2. The DNS server finds the IP address for google.com.
    3. The request travels through the router and ISP across the internet.
    4. The request reaches a web server in a data center.
    5. The server sends the webpage data back to the browser.

Flow diagram

User Device
(Laptop / Phone)
|
▼
Home Router
(Home Network)
|
▼
ISP

(Internet Service Provider)
|
▼
Internet Backbone
|
▼
Data Center
|
▼
Web Server
|
▼
Response Sent Back to User

Diagram Description

When a user enters a website address, the request travels from the user device through the router and ISP across the internet to a web server in a data center, which processes the request and sends the webpage back to the user.

13. **Why do organizations move from on-premises infrastructure to cloud environments? List at least three reasons.**

Reason	**Business Benefit (Security Focus)**
Cost savings	Pay-per-use eliminates hardware, maintenance, power costs.
Scalability	Auto-scale resources up/down instantly for demand spikes.
Disaster recovery	Built-in redundancy, automated backups reduce RTO/RPO.

14. **Explain the difference between Public cloud, Private cloud, and Hybrid cloud.**

Cloud Type	**Description**
Public	Services provided over the internet
Private	Infrastructure dedicated to one organization
Hybrid	Combination of public and private cloud

15. **If a company stores medical records in the cloud, what security concerns should they think about?**

A company storing medical records in the cloud must consider several security risks.

- Data privacy: Protect patient information from unauthorized access.
- Data breaches: Prevent attackers from stealing sensitive health records.

- Access control: Ensure only authorized staff can view medical data.
- Compliance: Follow healthcare regulations such as HIPAA.
- Data encryption: Protect records during storage and transmission.

Part C – GRC Connection (Foundation to RMF)

16. **Why is understanding how computers and networks work important for Governance, Risk, and Compliance (GRC)?**

    Understanding technology helps GRC professionals identify risks, evaluate security controls, and ensure systems meet regulatory and security requirements.

17. **Identify three risks related to Routers, Data centers, and Cloud environments.**

Infrastructure	Risk 1	Risk 2	Risk 3
Routers	Default credentials unchanged	Unpatched firmware vulns	Exposed management interface
Data Centers	Single power grid failure	HVAC/cooling system failure	Physical access control fail
Cloud	Misconfigured S3 buckets	No MFA on IAM accounts	Shared responsibility gaps

18. **What could happen if a company's switch fails in a corporate office? Explain the business impact.**

    If a company's network switch fails, devices in the office cannot communicate with each other or access the network.

    Business impacts may include:

    - Employees lose access to the internet and internal systems
    - File servers and shared resources become unavailable
    - Communication tools such as email or VoIP may stop working
    - Business operations may be delayed or interrupted

19. **Explain the difference between Data, Information, and Assets. Why are they important in risk management?**

Term	Description
Data	Raw facts or numbers without context
Information	Processed data that has meaning

Assets	Valuable resources that an organization must protect

20. **Imagine you are a security analyst. A company wants to migrate to the cloud. List 5 questions you would ask before approving the migration.**

    1. Which cloud service model will be used (IaaS, PaaS, or SaaS)?
    2. What data classification level will be stored in the cloud (public, internal, confidential)?
    3. Does the cloud provider support regulatory compliance requirements such as HIPAA or PCI?
    4. Where will the data be stored geographically, and does it meet data residency regulations?
    5. How will identity and access management (IAM) and multi-factor authentication (MFA) be implemented?

Create a one-page visual cheat sheet explaining:

- Computer components
- Networking devices
- Cloud models

Please see the next page below for the visual cheat sheet.

COMPUTER, NETWORKING, AND CLOUD FUNDAMENTALS

Cybersecurity Foundations Cheat Sheet

COMPUTER COMPONENTS

Input

Devices that send data into a computer

Examples:

- Keyboard
- Mouse
- Microphone
- Camera

Processing

CPU (Central Processing Unit)

- Executes programs
- Processes data
- Controls operations

Key Idea:
The CPU is the brain of the computer

Storage

Where data & programs are kept

HDD (Hard Disk Drive)

- Spinning disks
- Large capacity

SSD (Solid State Drive)

- Faster & more reliable
- No moving parts

Cloud Storage

- Data stored on remote servers
- Accessed via Internet

Output

Devices that **display** results

Examples:

- Monitor
- Printer
- Speakers

NETWORKING DEVICES

Router

Connects **network to the Internet**

- Directs traffic between networks
- Connects home/office to Internet

Switch

Connects devices within **same network**

- Links computers together
- Sends data to correct device

Modem

Translates signals ↔ Network & ISP

- Digital → ISP signal
- ISP signal → Digital

Server

Provides services & data to other computers

Examples:

- Website hosting
- File storage
- Databases

CLOUD COMPUTING MODELS

SaaS – Software as a Service

Applications delivered through the Internet

Examples:

- Gmail
- Microsoft 365
- Salesforce

Key Idea:
You simply log in and use the **software**

PaaS – Platform as a Service

Platform to build & deploy apps

Examples:

- Google App Engine
- Heroku

Key Idea:
Developers focus on building applications

IaaS – Infrastructure as a Service

Virtual servers & **infrastructure**

Examples:

- Amazon EC2
- Microsoft Azure VMs

Key Idea:
You rent virtual servers

Modern Organsizations Rely On:

- **Computers** → Process Data
- **Networks** → Move Data
- **Cloud Services** → Store & Scale Data